Multiplying Bigger Numbers

Three-digit times two-digit: six pieces, or two rows.

243 × 36: six pieces

	200	40	3
30	6,000	1,200	90
6	1,200	240	18

Hundreds, tens and ones across the top; tens and ones down the side.

6,000 + 1,200 + 90 + 1,200 + 240 + 18 = 8,748

The same in two rows

$$\begin{array}{r} 243 \\ \times \quad 36 \\ \hline 1{,}458 \quad (\times 6) \\ 7{,}290 \quad (\times 30) \\ \hline 8{,}748 \end{array}$$

Row 1: 243 × 6 = 1,458 (the bottom pieces).
Row 2: 243 × 30 = 7,290 (the top pieces).
243 × 30 is 243 × 3 tens = 729 tens.
Add the two rows: 8,748.

Your turn!

312 × 23 = 7,176

For teachers

Goal: multiply three- and four-digit numbers by two-digit numbers (5.NBT.5). The two rows are the area model's pieces added in groups. Say “243 times 3 tens is 729 tens”, never “put down a zero”.

Multiplying Bigger Numbers: Practice

Split, multiply each piece, add.

Part 1 Hundreds times tens 3 hundreds × 4 tens = 12 thousands.

1) 300 × 40 = 12,000	2) 200 × 60 = 12,000	3) 500 × 70 = 35,000
4) 400 × 30 = 12,000	5) 700 × 20 = 14,000	6) 600 × 90 = 54,000
7) 800 × 50 = 40,000	8) 900 × 40 = 36,000	9) 200 × 80 = 16,000
10) 300 × 70 = 21,000	11) 500 × 50 = 25,000	12) 400 × 60 = 24,000

Multiplying Bigger Numbers: Practice (continued)

Part 2 Six pieces The parts are on the outside. Fill in each piece.

13) **234 × 15 = 3,510**

	200	30	4
10	2,000	300	40
5	1,000	150	20

2,000 + 300 + 40 + 1,000 + 150 + 20 = 3,510

14) **162 × 37 = 5,994**

	100	60	2
30	3,000	1,800	60
7	700	420	14

3,000 + 1,800 + 60 + 700 + 420 + 14 = 5,994

15) **425 × 23 = 9,775**

	400	20	5
20	8,000	400	100
3	1,200	60	15

8,000 + 400 + 100 + 1,200 + 60 + 15 = 9,775

Multiplying Bigger Numbers: Practice (continued)

Part 3 Split it yourself Write the parts on the outside, then fill in each piece.

16) **318 × 24 = 7,632**

	300	10	8
20	6,000	200	160
4	1,200	40	32

6,000 + 200 + 160 + 1,200 + 40 + 32 = 7,632

17) **256 × 43 = 11,008**

	200	50	6
40	8,000	2,000	240
3	600	150	18

8,000 + 2,000 + 240 + 600 + 150 + 18 = 11,008

18) **173 × 52 = 8,996**

	100	70	3
50	5,000	3,500	150
2	200	140	6

5,000 + 3,500 + 150 + 200 + 140 + 6 = 8,996

Multiplying Bigger Numbers: Practice (continued)

Part 4 Two rows Row 1: times the ones. Row 2: times the tens. Then add.

19)
$$\begin{array}{r} 214 \\ \times \quad 32 \\ \hline 428 \quad (\times 2) \\ 6{,}420 \quad (\times 30) \\ \hline 6{,}848 \end{array}$$

20)
$$\begin{array}{r} 365 \\ \times \quad 24 \\ \hline 1{,}460 \quad (\times 4) \\ 7{,}300 \quad (\times 20) \\ \hline 8{,}760 \end{array}$$

21)
$$\begin{array}{r} 128 \\ \times \quad 57 \\ \hline 896 \quad (\times 7) \\ 6{,}400 \quad (\times 50) \\ \hline 7{,}296 \end{array}$$

22)
$$\begin{array}{r} 432 \\ \times \quad 18 \\ \hline 3{,}456 \quad (\times 8) \\ 4{,}320 \quad (\times 10) \\ \hline 7{,}776 \end{array}$$

23)
$$\begin{array}{r} 251 \\ \times \quad 46 \\ \hline 1{,}506 \quad (\times 6) \\ 10{,}040 \quad (\times 40) \\ \hline 11{,}546 \end{array}$$

24)
$$\begin{array}{r} 617 \\ \times \quad 35 \\ \hline 3{,}085 \quad (\times 5) \\ 18{,}510 \quad (\times 30) \\ \hline 21{,}595 \end{array}$$

25)
$$\begin{array}{r} 183 \\ \times \quad 29 \\ \hline 1{,}647 \quad (\times 9) \\ 3{,}660 \quad (\times 20) \\ \hline 5{,}307 \end{array}$$

26)
$$\begin{array}{r} 526 \\ \times \quad 14 \\ \hline 2{,}104 \quad (\times 4) \\ 5{,}260 \quad (\times 10) \\ \hline 7{,}364 \end{array}$$

27)
$$\begin{array}{r} 342 \\ \times \quad 67 \\ \hline 2{,}394 \quad (\times 7) \\ 20{,}520 \quad (\times 60) \\ \hline 22{,}914 \end{array}$$

Part 5 Multiply No help this time.

28) $326 \times 21 = 6{,}846$

29) $147 \times 38 = 5{,}586$

Multiplying Bigger Numbers: Practice (continued)

Part 5 Multiply (continued)

30) $512 \times 46 =$ **23,552**

31) $238 \times 17 =$ **4,046**

32) $403 \times 25 =$ **10,075**

33) $671 \times 34 =$ **22,814**

34) $295 \times 42 =$ **12,390**

35) $184 \times 63 =$ **11,592**

36) $736 \times 15 =$ **11,040**

37) $459 \times 27 =$ **12,393**

38) $128 \times 74 =$ **9,472**

39) $387 \times 56 =$ **21,672**

40) $604 \times 39 =$ **23,556**

41) $215 \times 88 =$ **18,920**

Multiplying Bigger Numbers: Practice (continued)

Part 5 Multiply (continued)

42) $548 \times 23 =$ **12,604**

43) $963 \times 12 =$ **11,556**

44) $372 \times 45 =$ **16,740**

45) $819 \times 26 =$ **21,294**

46) $256 \times 91 =$ **23,296**

47) $437 \times 68 =$ **29,716**

Part 6 Four digits times one digit No help this time.

48) $2{,}134 \times 3 =$ **6,402**

49) $4{,}206 \times 7 =$ **29,442**

50) $1{,}573 \times 4 =$ **6,292**

51) $3{,}058 \times 6 =$ **18,348**

52) $6{,}125 \times 8 =$ **49,000**

53) $2{,}947 \times 5 =$ **14,735**

Multiplying Bigger Numbers: Practice (continued)

Part 6 Four digits times one digit (continued)

54) $5{,}312 \times 9 =$ **47,808**

55) $1{,}806 \times 4 =$ **7,224**

56) $7{,}243 \times 3 =$ **21,729**

57) $3{,}615 \times 6 =$ **21,690**

Part 7 Multiply

58) $672 \times 35 =$ **23,520**

59) $763 \times 43 =$ **32,809**

60) $728 \times 41 =$ **29,848**

61) $163 \times 54 =$ **8,802**

62) $353 \times 47 =$ **16,591**

63) $182 \times 39 =$ **7,098**

64) $672 \times 59 =$ **39,648**

65) $236 \times 54 =$ **12,744**

Multiplying Bigger Numbers: Practice (continued)

Part 7 Multiply (continued)

66) $526 \times 16 =$ 8,416

67) $162 \times 15 =$ 2,430

68) $621 \times 38 =$ 23,598

69) $113 \times 29 =$ 3,277

70) $873 \times 19 =$ 16,587

71) $113 \times 22 =$ 2,486

72) $194 \times 12 =$ 2,328

73) $664 \times 59 =$ 39,176

74) $153 \times 33 =$ 5,049

75) $542 \times 49 =$ 26,558

76) $371 \times 26 =$ 9,646

77) $736 \times 48 =$ 35,328

Multiplying Bigger Numbers: Practice (continued)

Part 7 Multiply (continued)

78) $358 \times 17 =$ 6,086

79) $848 \times 59 =$ 50,032

80) $237 \times 38 =$ 9,006

81) $275 \times 42 =$ 11,550

82) $344 \times 58 =$ 19,952

83) $355 \times 48 =$ 17,040

84) $148 \times 17 =$ 2,516

85) $464 \times 12 =$ 5,568

86) $497 \times 24 =$ 11,928

87) $527 \times 55 =$ 28,985

88) $459 \times 14 =$ 6,426

89) $604 \times 24 =$ 14,496

Multiplying Bigger Numbers: Practice (continued)

Part 7 Multiply (continued)

90) $182 \times 36 =$ **6,552**

91) $883 \times 43 =$ **37,969**

92) $121 \times 13 =$ **1,573**

93) $165 \times 47 =$ **7,755**

Part 8 Remember? Problems from earlier packets.

94) $3.6 + 2.75 =$ **6.35**

95) $8.2 - 4.67 =$ **3.53**

96) 4.28 to the nearest tenth = **4.3**

97) 0.45 **>** 0.405

98) in 3.142 the 4 is worth **0.04**

99) $2.7 \times 1{,}000 =$ **2,700**

Multiplying Bigger Numbers: Practice (continued)

Part 9 Word problems Split, multiply, add.

100) A school bus holds 48 kids. How many kids can 125 buses hold?

Number sentence **125 × 48 = 6,000**

Answer **6,000** kids

Draw or show your work

101) A farm packs 236 boxes of eggs. Each box has 12 eggs. How many eggs is that?

Number sentence **236 × 12 = 2,832**

Answer **2,832** eggs

Draw or show your work

102) A plane flies 465 miles each day for 21 days. How far does it fly?

Number sentence **465 × 21 = 9,765**

Answer **9,765** miles

Draw or show your work

103) A box of paper has 1,250 sheets. How many sheets are in 8 boxes?

Number sentence **1,250 × 8 = 10,000**

Answer **10,000** sheets

Draw or show your work

104) Challenge: A theater has 34 rows of 28 seats. 875 seats are full. How many are empty?

Number sentence **34 × 28 = 952, 952 − 875 = 77**

Answer **77** seats

Draw or show your work

Check-In: Multiplying Bigger Numbers

Show what you know. No help this time.

Mastery: 8 of 10 (80%)

Part 1 Hundreds times tens

1) $300 \times 90 =$ **27,000**

2) $800 \times 90 =$ **72,000**

3) $700 \times 70 =$ **49,000**

Part 2 Two rows

4)

$$\begin{array}{r} 418 \\ \times \quad 23 \\ \hline 1{,}254 \quad (\times 3) \\ 8{,}360 \quad (\times 20) \\ \hline 9{,}614 \end{array}$$

5)

$$\begin{array}{r} 152 \\ \times \quad 47 \\ \hline 1{,}064 \quad (\times 7) \\ 6{,}080 \quad (\times 40) \\ \hline 7{,}144 \end{array}$$

6)

$$\begin{array}{r} 324 \\ \times \quad 26 \\ \hline 1{,}944 \quad (\times 6) \\ 6{,}480 \quad (\times 20) \\ \hline 8{,}424 \end{array}$$

Part 3 Multiply

7) $382 \times 19 =$ **7,258**

8) $527 \times 34 =$ **17,918**

9) $243 \times 17 =$ **4,131**

10) $275 \times 36 =$ **9,900**